

Modelling & Simulation of The Magic Man

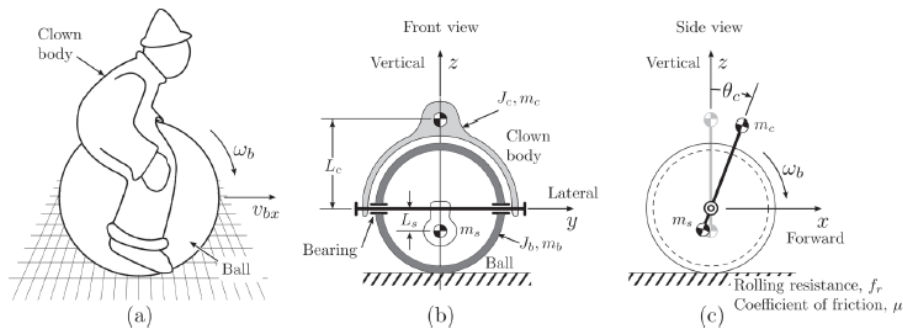


Figure B-8.32 (a) Magic man. (b) Model schematic. (c) Motion of clown body relative to rolling ball as balanced pendula.

- For the development of the model and bond graph, I will be using the Lagrangian Formulation.
- This model has a holonomic constraint: It is only rotates about the y axis and moves in the x-z plane
- Key Elements:
 1. Mass of the clown (m_c) with Length L_1
 2. Mass of the Counter weight (m_s) with length L_2
 3. mass of the ball (m_b) with radius R

Note: $z=0$ is the ground

- Generalized Coordinates are (x, θ_c) : These coordinates are able to define the system fully for all key elements in the system

• Steps for $\mathcal{L}$ Formulation

Kinetic Energy: $T_{\dot{x}, \dot{\theta}_c} = \frac{m_b \dot{x}^2}{2} + \frac{J_b \dot{\theta}^2}{2} + \frac{m_c v_c^2}{2} + \frac{m_s v_s^2}{2}$

$\dot{x} = R \omega_b$

No. Slip

$\dot{\theta} = \omega_b = \dot{x}/R$

$v_c = \dot{\vec{r}}_c \cdot \dot{\vec{r}}_c$

$\vec{r}_c = (x + L_1 \sin \theta_c) \hat{i} + (R + L_1 \cos \theta_c) \hat{j}$

$\therefore \dot{\vec{r}}_c = (\dot{x} + L_1 \cos \theta_c \dot{\theta}_c) \hat{i} - L_1 \sin \theta_c \dot{\theta}_c \hat{j}$

$$\therefore v_c^2 = (\dot{x} + l_1 \cos \theta_c \dot{\theta}_c)^2 + (-l_1 \sin \theta_c \dot{\theta}_c)^2 = \dot{x}^2 + 2\dot{x}l_1 \cos \theta_c \dot{\theta}_c + l_1^2 \cos^2 \theta_c \dot{\theta}_c^2 + l_1^2 \sin^2 \theta_c \dot{\theta}_c^2$$

$$\sin^2 + \cos^2 = 1$$

$$\therefore$$

$$\boxed{v_c^2 = \dot{x}^2 + 2\dot{x}l_1 \cos \theta_c \dot{\theta}_c + l_1^2 \dot{\theta}_c^2}$$

$$v_s^2 = \dot{r}_s^2 \cdot \dot{r}_s^2$$

$$\vec{r}_s = (x - l_2 \sin \theta_c) \hat{i} + (R - l_2 \cos \theta_c) \hat{j}$$

$$\dot{\vec{r}}_s = (\dot{x} - l_2 \cos \theta_c \dot{\theta}_c) \hat{i} + (l_2 \sin \theta_c \dot{\theta}_c) \hat{j}$$

$$\boxed{v_s^2 = \dot{x}^2 - 2\dot{x}l_2 \cos \theta_c \dot{\theta}_c + l_2^2 \dot{\theta}_c^2}$$

$$\therefore T_{\dot{x}\dot{\theta}_c} = \frac{m_b \dot{x}^2}{2} + \frac{J_b \dot{\theta}_c^2}{2} + \frac{m_c v_c^2}{2} + \frac{m_s v_s^2}{2}$$

$$T_{\dot{x}\dot{\theta}_c} = \frac{(m_b + m_c + m_s) \dot{x}^2}{2} + (m_c l_1 - m_s l_2) \cos \theta_c \dot{x} \dot{\theta}_c + \frac{(m_c l_1^2 + m_s l_2^2) \dot{\theta}_c^2}{2} + \frac{J_b \dot{x}^2}{2R^2}$$

$$\text{Generalized momenta: } p_x = \frac{\partial T_{\dot{x}\dot{\theta}_c}}{\partial \dot{x}} = (m_b + m_c + m_s) \dot{x} + (m_c l_1 - m_s l_2) \cos \theta_c \dot{\theta}_c + \frac{J_b \dot{x}}{R^2}$$

$$p_{\theta_c} = \frac{\partial T_{\dot{x}\dot{\theta}_c}}{\partial \dot{\theta}_c} = (m_c l_1 - m_s l_2) \cos \theta_c \dot{x} + (m_c l_1^2 + m_s l_2^2) \dot{\theta}_c$$

$$\begin{bmatrix} p_x \\ p_{\theta_c} \end{bmatrix} = \begin{bmatrix} (m_b + m_c + m_s) + \frac{J_b}{R^2} & (m_c l_1 - m_s l_2) \cos \theta_c \\ (m_c l_1 - m_s l_2) \cos \theta_c & (m_c l_1^2 + m_s l_2^2) \end{bmatrix} \begin{bmatrix} \dot{x} \\ \dot{\theta}_c \end{bmatrix}$$

$$\text{Potential Energy: } U_{x\theta_c} = m_c g z_c + m_s g z_s = m_c g (R + l_1 \cos \theta_c) + m_s g (R - l_2 \cos \theta_c)$$

$$U_{x\theta_c} = gR(m_c + m_s) + g(m_c l_1 - m_s l_2) \cos \theta_c$$

Dissipation: - I assumed no slipping, but other sources of dissipation are

- Rolling resistance in x direction: $F_r = f_r W = f_r (m_c + m_s + m_b) g = f_r W$

$$\therefore Q_x = \frac{-T_{rr}}{R} = -f_r W \tanh\left(\frac{\dot{x}}{R\omega_0}\right)$$

- Bearing Friction: Friction in the bearing where the body rotates

$$Q_{\theta_c} = -T_f \tanh\left(\frac{\dot{\theta}_c}{\omega_0}\right) - B \dot{\theta}_c - T_{ao}$$

Lagrange Terms:

if $M = (m_b + m_c + m_s + \frac{J_b}{R^2})$ & $K = (m_c l_1 - m_s l_2)$ $I = (m_c l_1^2 + m_s l_2^2)$

$$\therefore \mathcal{L} = T - U = \left(\frac{1}{2} M \dot{x}^2 + K \cos \theta_c \dot{x} \dot{\theta}_c + \frac{1}{2} I \dot{\theta}_c^2 \right) - \left(g K \cos \theta_c + g R (m_c + m_s) \right)$$

$$\frac{\partial T}{\partial \dot{x}} = M \dot{x} + K \cos \theta_c \dot{\theta}_c$$

$$\frac{\partial T}{\partial \dot{\theta}_c} = K \cos \theta_c \dot{x} + I \dot{\theta}_c$$

$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{x}} \right) = M \ddot{x} + K \cos \theta_c \ddot{\theta}_c - K \sin \theta_c \dot{\theta}_c^2$$

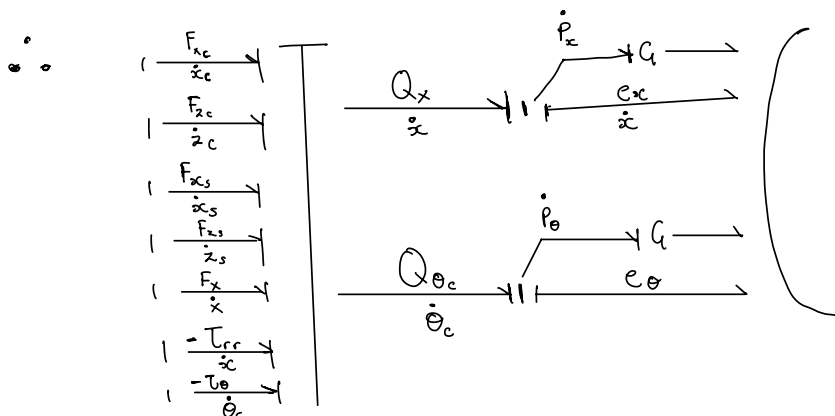
$$\frac{d}{dt} \left(\frac{\partial T}{\partial \dot{\theta}_c} \right) = K \cos \theta_c \ddot{x} - K \sin \theta_c \dot{x} \dot{\theta}_c + I \ddot{\theta}_c$$

$$\frac{\partial T}{\partial \theta_c} = -K \sin \theta_c \dot{x} \dot{\theta}_c \Rightarrow \frac{\partial \mathcal{L}}{\partial \theta_c} = \frac{\partial T}{\partial \theta_c} - \frac{\partial V}{\partial \theta_c} = -K \sin \theta_c \dot{x} \dot{\theta}_c + g K \sin \theta_c$$

$$\frac{\partial V}{\partial \theta_c} = -g K \sin \theta_c$$

$$\therefore M \ddot{x} + K \cos \theta_c \ddot{\theta}_c - K \sin \theta_c \dot{\theta}_c^2 = Q_x$$

$$K \cos \theta_c \ddot{x} + I \ddot{\theta}_c - g K \sin \theta_c = Q_{\theta_c}$$



if state equation are $[x, \theta_c, \dot{x}, \dot{\theta}_c]$ $x_1 = x$, $x_2 = \theta_c$, $x_3 = \dot{x}$, $x_4 = \dot{\theta}_c$

$$\dot{x}_1 = x_3, \quad \dot{x}_2 = x_4$$

$$\begin{bmatrix} M & k \cos x_2 \\ k \cos x_2 & I \end{bmatrix} \begin{bmatrix} \dot{x}_3 \\ \dot{x}_4 \end{bmatrix} = \begin{bmatrix} Q_x + k \sin x_2 x_4^2 \\ Q_\theta + g k \sin x_2 \end{bmatrix}$$

$$\text{determinant} = MI - (k \cos x_2)^2$$

$$\dot{x}_3 = \ddot{x} = \frac{I(Q_x + k \sin x_2 x_4^2) - (k \cos x_2)(Q_\theta + g k \sin x_2)}{(MI - (k \cos x_2)^2)}$$

$$\dot{x}_4 = \ddot{\theta}_c = \frac{-(k \cos x_2)(Q_x + k \sin x_2 x_4^2) + M(Q_\theta + g k \sin x_2)}{(MI - (k \cos x_2)^2)}$$

Simulation:

- Code: Before plot because I change friction values and initial condition to get a clear picture and would like to discuss that

```

1 import numpy as np
2 from scipy.integrate import solve_ivp
3 import matplotlib.pyplot as plt
4
5 # =====
6 # 1. Physical constants
7 # =====
8 g = 9.81 # gravity [m/s^2]
9
10 # =====
11 # 2. Given problem data
12 # =====
13 M_total = 0.2 # total mass [kg]
14 m_b = 0.1 # ball mass [kg]
15 R = 0.09 # ball radius [m]
16 mu = 0.3 # ground friction coefficient
17 v0 = 1 # initial forward velocity [m/s]
18
19 # =====
20 # 3. Estimated body masses
21 # =====
22 m_body = M_total - m_b
23 m_c = 0.12 * m_body # upper mass [kg]
24 m_s = m_body - m_c # lower mass [kg]
25
26 # =====
27 # 4. Geometry estimates
28 # =====
29 l1 = (1/4) * R # distance to upper mass [m]
30 l2 = 3 * l1 * (m_c / m_s) # distance to lower mass [m]
31
32 # =====
33 # 5. Ball inertia (thin-walled hollow sphere)
34 # =====
35 Jb = (2/3) * m_b * R**2
36
37 # =====
38 # 6. Reduced Lagrangian coefficients
39 # =====
40 M = m_b + m_c + m_s + Jb / R**2
41 K = (m_c * l1 - m_s * l2)
42 I = (m_c * l1**2 + m_s * l2**2)
43
44 # Normal force (no hopping assumed)
45 N = M_total * g
46
47 # =====
48 # 7. Dissipation parameters (tunable)
49 # =====
50 fr = 0.001 # rolling resistance coefficient
51 omega0 = 0.05 # smoothing speed for rolling resistance
52
53 tau_f0 = 0.001 # Coulomb bearing friction torque [Nm]
54 B = 0.0 # viscous bearing damping [Nm/s/rad]
55 tau_a0 = 0.0 # bias torque
56 omega_so = 0.05 # smoothing speed for bearing friction
57
58 # =====
59 # 8. State equations
60 # State: X = [x, theta_c, xdot, thetadot]
61 # =====
62 def dynamics(t, X):
63     x, theta, xdot, thetadot = X
64
65     # =====
66     # Dissipation forces
67     # =====
68
69     # Rolling resistance (acts in x)
70     Qx = -fr * N * np.tanh(xdot / (R * omega0))
71
72     # Bearing friction (acts in theta)
73     Qtheta = (
74         -tau_f0 * np.tanh(thetadot / omega_so)
75         - B * thetadot
76         - tau_a0
77     )
78
79     # Mass matrix determinant
80     # =====
81     Delta = M * I - (K * np.cos(theta))**2
82
83     # Right-hand side terms
84     # =====
85     rhs_x = Qx + K * np.sin(theta) * thetadot**2
86     rhs_theta = Qtheta + g * K * np.sin(theta)
87
88     # Accelerations
89     # =====
90     xddot = (I * rhs_x - K * np.cos(theta) * rhs_theta) / Delta
91     thetaddot = (-K * np.cos(theta) * rhs_x + M * rhs_theta) / Delta
92
93     return [xdot, thetadot, xddot, thetaddot]
94
95 # =====
96 # 9. Initial conditions
97 # =====
98 x0 = [
99     0.0, # x(0)
100     0.0, # theta_c(0)
101     v0, # xdot(0)
102     0, # thetadot(0)
103 ]

```

```

100 # -----
101 # 10. Time span and integration
102 # -----
103 t_span = (0, 80)
104 t_eval = np.linspace(0, 80, 3000)
105
106 sol = solve_ivp(
107     dynamics,
108     t_span,
109     x0,
110     t_eval=t_eval,
111     rtol=1e-6,
112     atol=1e-8
113 )
114
115 # -----
116 # 11. Extract results
117 # -----
118 t = sol.t
119 x = sol.y[0]
120 theta = sol.y[1]
121 xdot = sol.y[2]
122 thetadot = sol.y[3]
123
124 # -----
125 # 12. Plot results
126 # -----
127 plt.figure(figsize=(12, 9))
128
129 plt.subplot(4, 1, 1)
130 plt.plot(t, x)
131 plt.ylabel("x [m]")
132 plt.grid()
133
134 plt.subplot(4, 1, 2)
135 plt.plot(t, xdot)
136 plt.ylabel("xdot [m/s]")
137 plt.grid()
138
139 plt.subplot(4, 1, 3)
140 plt.plot(t, theta * 180 / np.pi)
141 plt.ylabel("theta [deg]")
142 plt.grid()
143
144 plt.subplot(4, 1, 4)
145 plt.plot(t, thetadot)
146 plt.ylabel("thetadot [rad/s]")
147 plt.xlabel("time [s]")
148 plt.grid()
149
150 plt.tight_layout()
151 plt.show()
152

```

Plots:

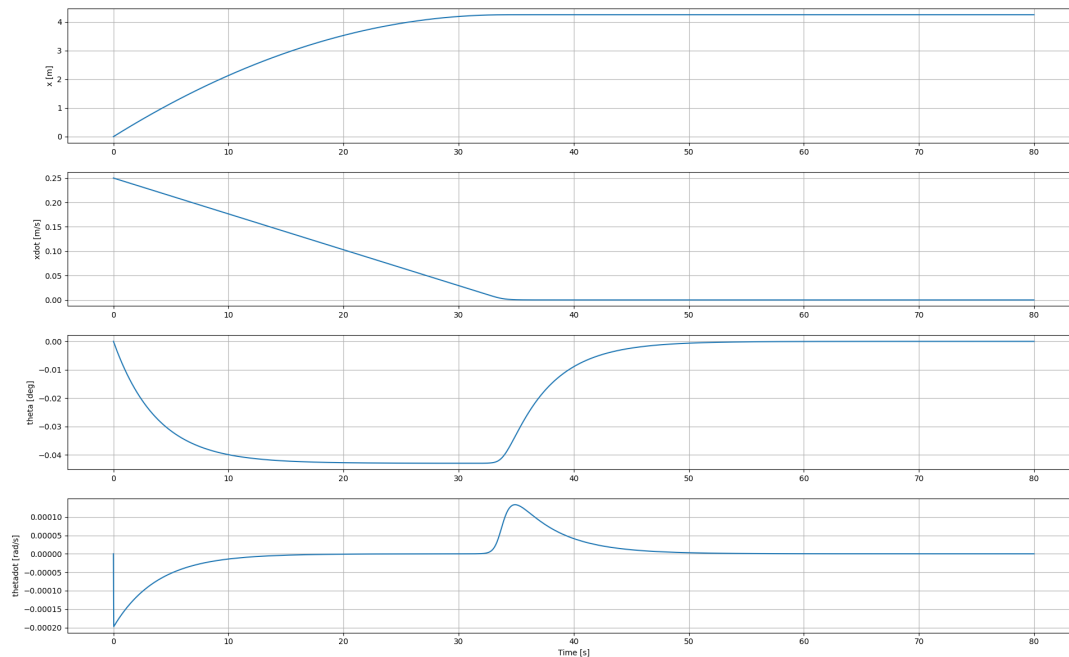
@ Initial Condition : $x = 0$ $\theta_c = 0$ $\dot{x} = 0.25$ $\dot{\theta}_c = 0$

```

# -----
# 7. Dissipation parameters (tunable)
# -----
fr = 0.001 # rolling resistance coefficient
omega0 = 0.05 # smoothing speed for rolling resistance

tau_f0 = 0.001 # Coulomb bearing friction torque [Nm]
B = 0.0 # viscous bearing damping [Nms/rad]
tau_a0 = 0.0 # bias torque
omega_so = 0.05 # smoothing speed for bearing friction

```



- In this result is consistent with the Lagrangian formulation constructed
- To be honest I expected a wave like theta graph but I believe it is the rotational resistance that prevents quick back and forth change in θ_c
- I also believe the sharp dip in $\dot{\theta}_c$ graph is very realistic, the clown body in a way counteracts the "sudden velocity". it goes back to the $\dot{\theta}_c = 0$ initial condition but rises again when the ball stopped to adjust. It is like a car braking there is a sudden push forward.
- The result is very realistic, I am confident in it.

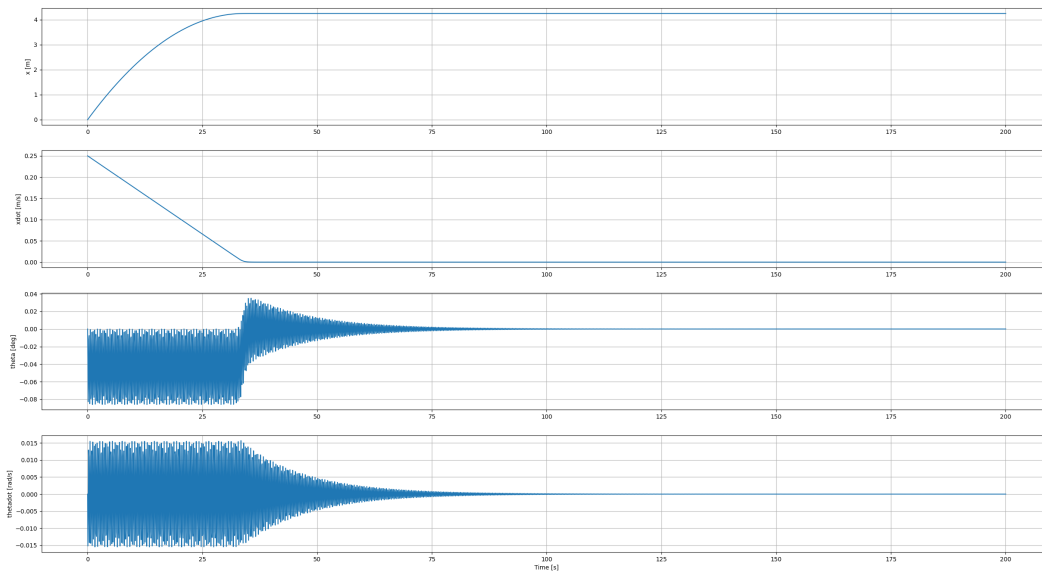
Plot 2: Same initial condition

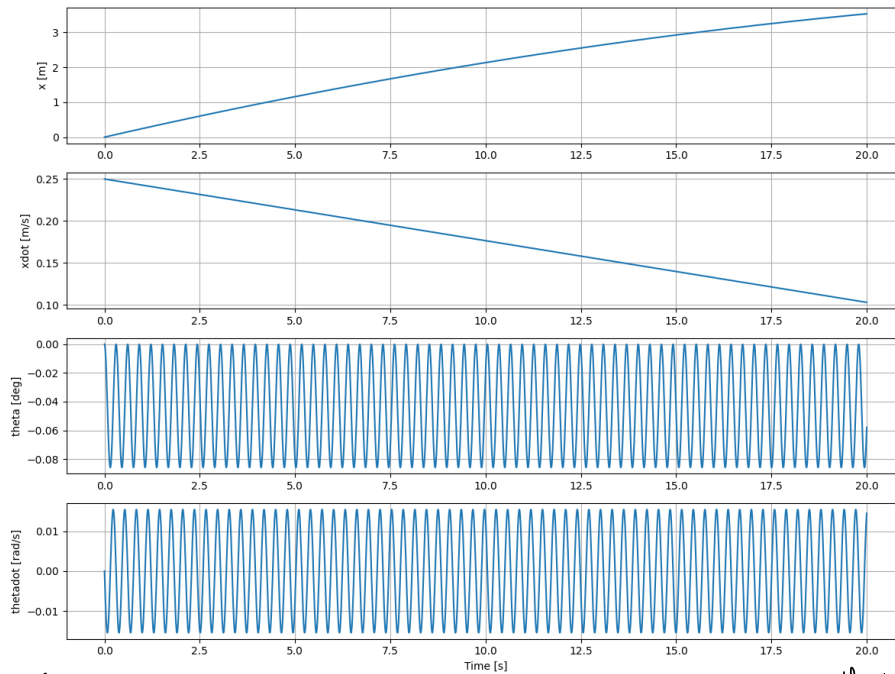
```

47 # =====
48 # 7. Dissipation parameters (tunable)
49 # =====
50 fr = 0.001 # rolling resistance coefficient
51 omega0 = 0.05 # smoothing speed for rolling resistance
52
53 tau_f0 = 0.0 # Coulomb bearing friction torque [Nm]
54 B = 0.0 # viscous bearing damping [Nms/rad]
55 tau_a0 = 0.0 # bias torque
56 omega_so = 0.05 # smoothing speed for bearing friction
57

```

no rotational resistance



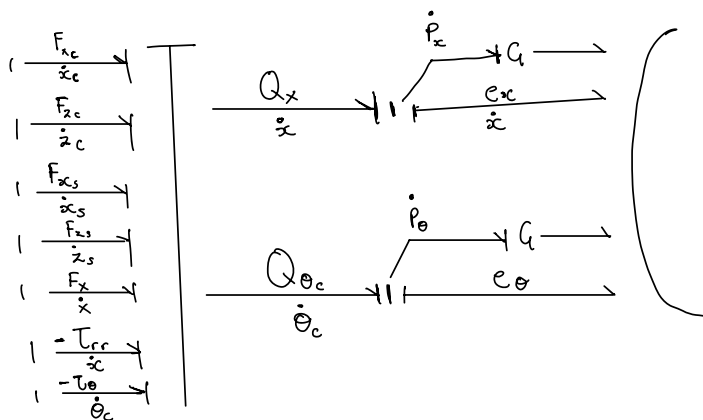


Close
look

- There is the wave like motion I was looking for in theta
- Notice a change in oscillation when the toy man stopped. it follows the same pattern as the last one

- Very confident in this model

* Discussion of Bond Graph



- The bond graph is really a reflection of the lagrange formulation
- Instead of using v_s & v_c as flow variables i used it components as x_c, z_c & x_s, z_s are dependent coordinates in the systems